SOFTWARE TESTING LAB

PERFORMANCE TESTING AND TEST SUMMARY REPORT

Practice Software Testing Toolshop

Course / Subject	ISWE305P – Software Testing Lab
Application Under Test	https://practicesoftwaretesting.com/
Performance Tool	Apache JMeter 5.6.3
Assigned Member	Jayaprakash — Member 3
Role	Performance Testing and Test Summary Report
Report Basis	Supplied JMeter screenshots, comparison document, SRS and functional testing report

Prepared using the evidence provided for the Software Testing Lab project.

1. Project Overview

This report summarizes the performance testing carried out on the public Practice Software Testing Toolshop homepage. Apache JMeter was used to send HTTP GET requests and record response-time, throughput and error-rate measurements. The supplied evidence contains three exploratory runs: a baseline run with 1 virtual user, a moderate run with 5 virtual users, and a higher-load run with 10 virtual users. These tests provide a limited comparison of homepage HTTP behaviour; they are not a capacity certification or a comprehensive stress test.

1.1 Objectives

- Measure the homepage HTTP response time under the supplied test configurations.
- Compare average, minimum and maximum response times, standard deviation, throughput and error percentage across the three runs.
- Present the observed performance trends in tables and charts.
- Summarize the results, limitations and recommendations without inventing missing metrics.
- Combine the performance evidence with the supplied functional-testing information for the final report.

1.2 Scope

The measured scope is the homepage HTTP request at path "/" on practicesoftwaretesting.com. The report covers the JMeter configuration and the three recorded runs described in JMeter_Results_Comparison.docx. It does not claim to measure Angular/browser rendering, visual layout time, user interaction latency, database internals or private backend behaviour.

2. Test Environment and Configuration

Parameter	Recorded configuration / evidence
Application	Practice Software Testing Toolshop homepage
Target host	practicesoftwaretesting.com
Protocol / method	HTTPS / HTTP GET
Request path	/
Tool version	Apache JMeter 5.6.3, as stated in the comparison document
Baseline run	1 virtual user; 1-second ramp-up; 5 loops/samples
Moderate run	5 virtual users; 25 recorded samples
Higher-load run	10 virtual users; 30 recorded samples
Total recorded samples	60 across the three runs
Safety note	Exploratory low-volume checks against a public demo; no stress-test certification

2.1 JMeter Configuration Evidence

The following screenshots supplied for the project are included as evidence. They show the baseline HTTP Request sampler, the Thread Group settings, the Aggregate Report, and HTTP Request Defaults. The configuration screenshots relate to the baseline setup; the comparison document supplies the summary metrics for all three runs.

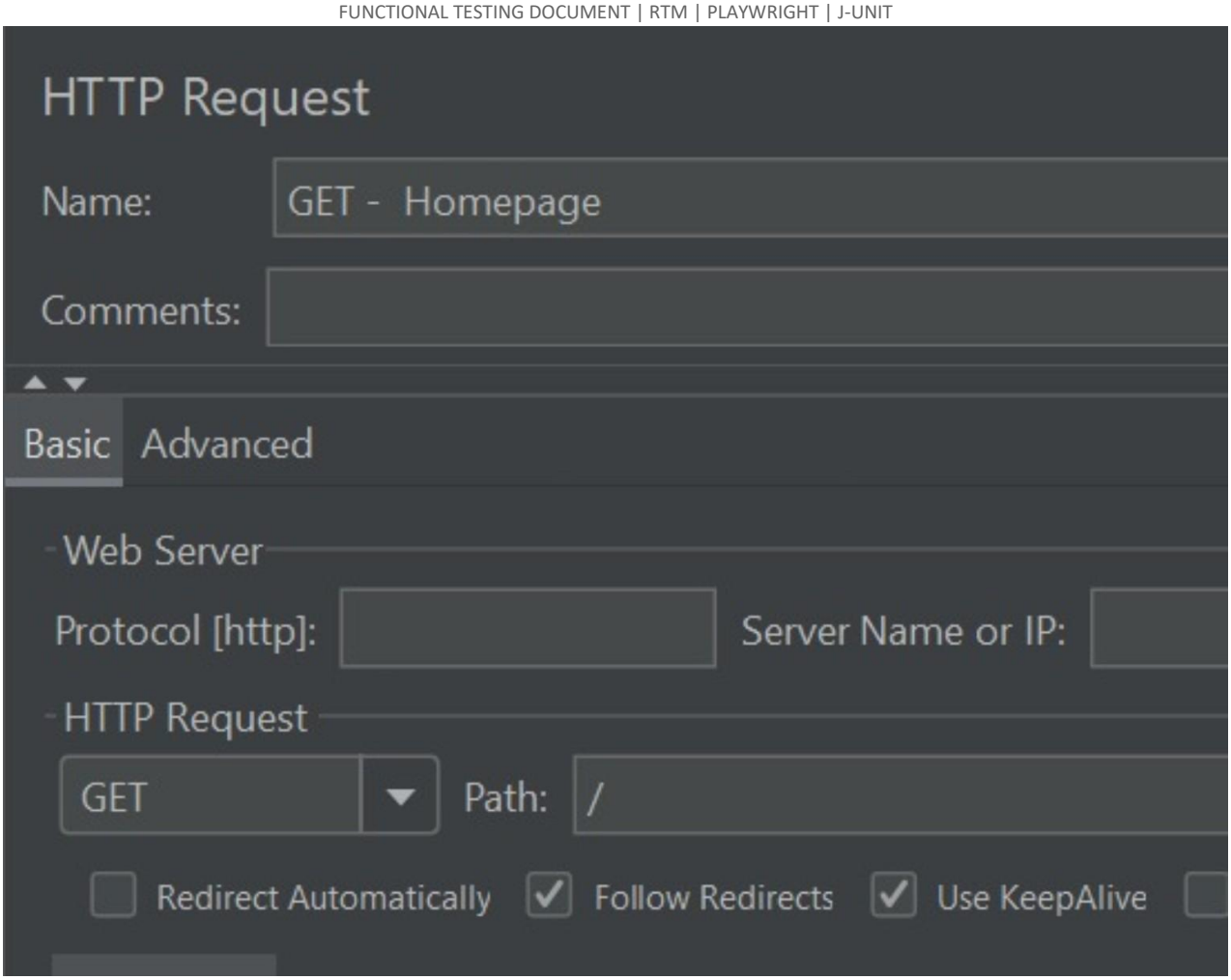


Figure 1. HTTP Request sampler — GET - Homepage, path "/".

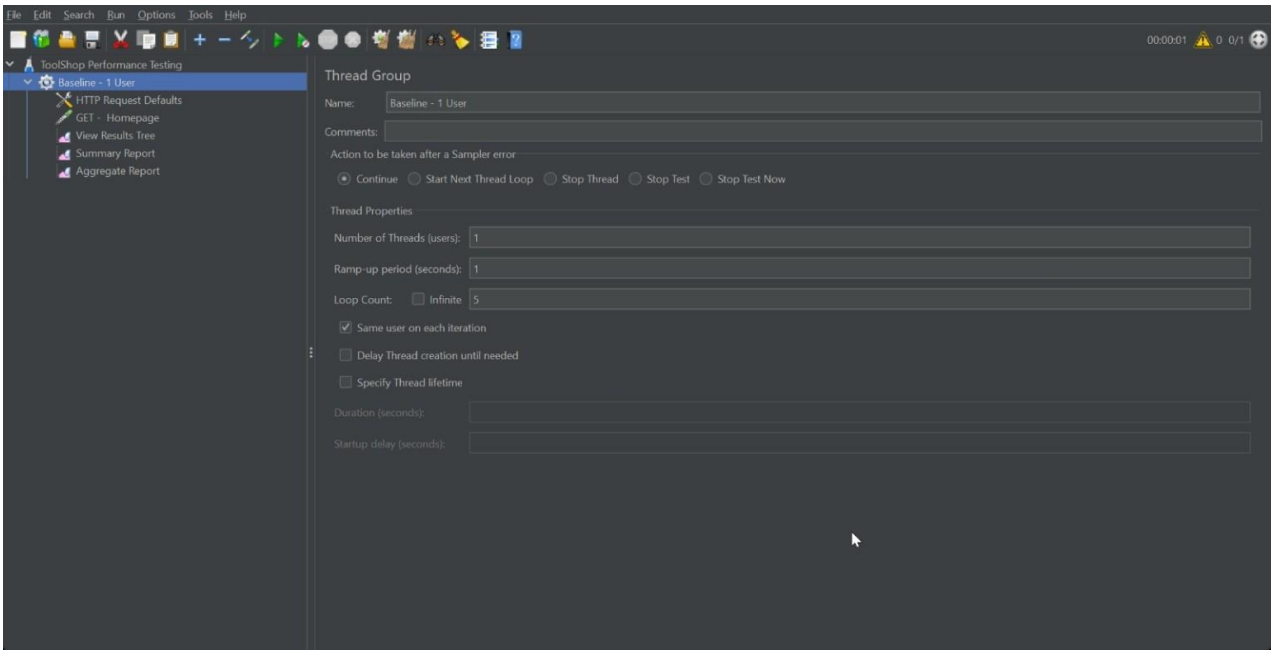


Figure 2. Thread Group — Baseline - 1 User, one user, one-second ramp-up and five loops.

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
GET - Homepage	5	318	219	681	181.44	0.00%	3.1/sec	31.38	0.39	10270.2
TOTAL	5	318	219	681	181.44	0.00%	3.1/sec	31.38	0.39	10270.2

Figure 3. Aggregate Report — baseline run measurements.

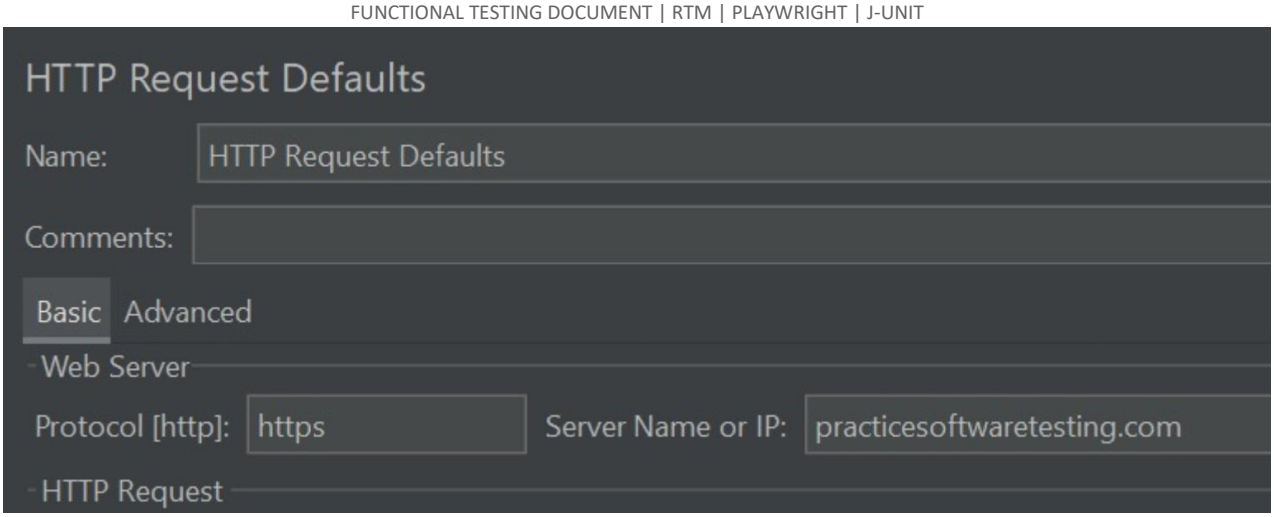


Figure 4. HTTP Request Defaults — HTTPS and practicesoftwaretesting.com.

3. JMeter Results — Comparison Table

The following values are reproduced from the supplied JMeter Results Comparison document. The clipped 5-user average-bytes value is left unestimated.

Metric	Baseline 1 user	Moderate 5 users	Higher load 10 users
Samples	5	25	30
Average response (ms)	318	394	458
Minimum response (ms)	219	244	239
Maximum response (ms)	681	1,010	958
Standard deviation (ms)	181.44	210.01	251.00
Error rate	0.00%	0.00%	0.00%
Throughput (req/sec)	3.1	4.2	2.9
Received (KB/sec)	31.38	41.26	28.81
Sent (KB/sec)	0.39	0.53	0.36
Average bytes/response	10,270.2	Not fully visible	10,197.2

Percentiles were visible only for the 10-user run: median 310 ms, 90th percentile 826 ms, 95th percentile 908 ms and 99th percentile 958 ms. These values are not compared across runs because equivalent percentile values were not visible for the other runs.

4. Charts and Performance Trends

The charts below are taken from the supplied JMeter_Results_Comparison.docx and retain its reported comparisons.

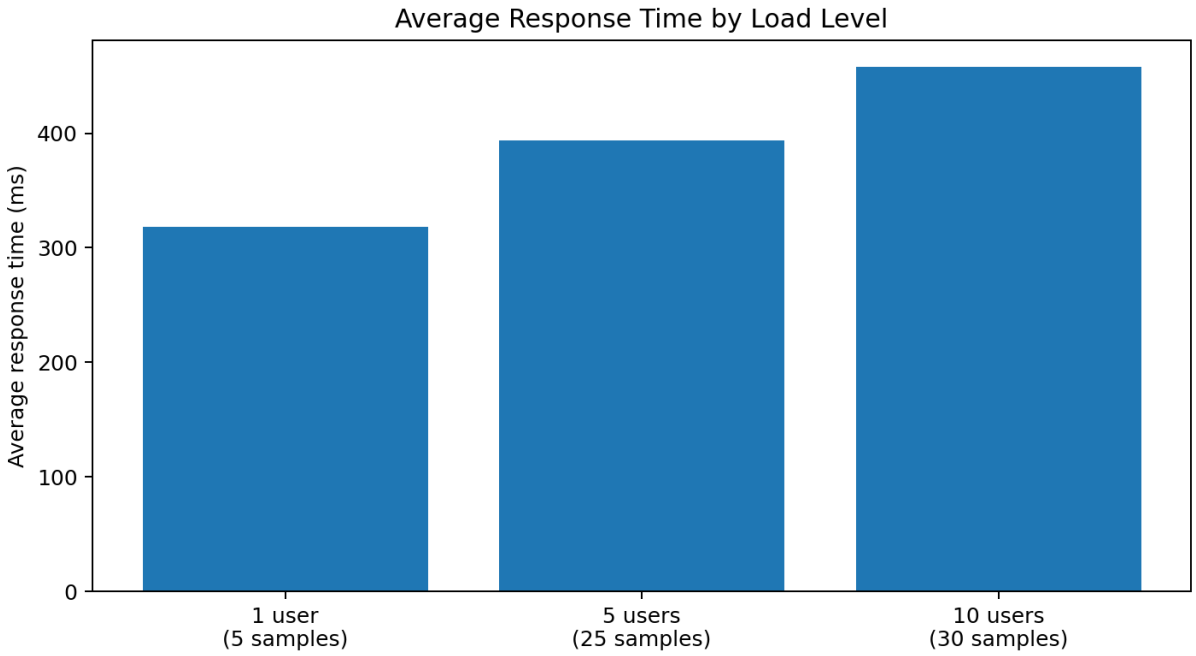


Figure 5. Average response time comparison.

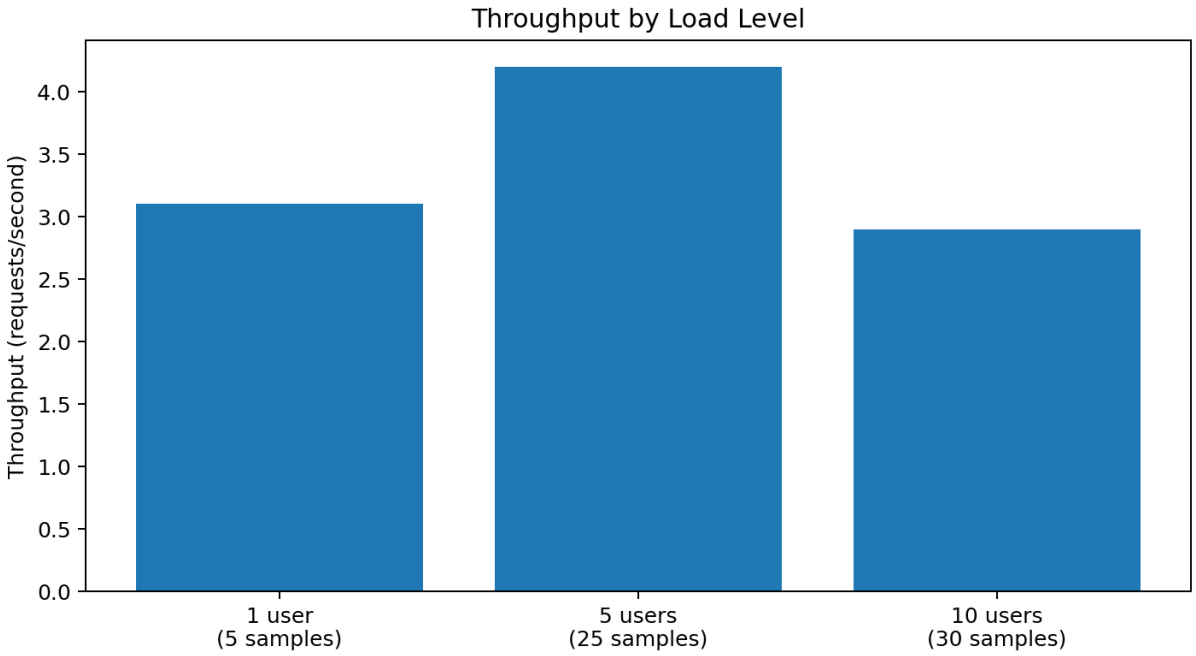


Figure 6. Throughput comparison.

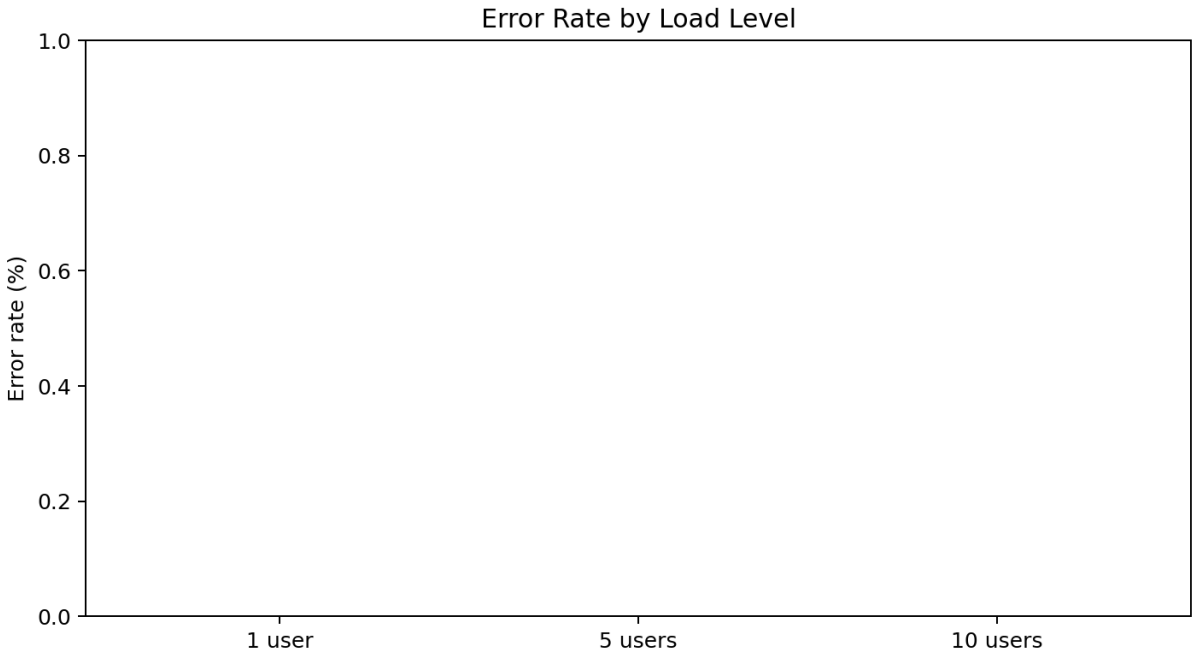


Figure 7. Error-rate comparison.

4.1 Average Response Time

Average response time increased from 318 ms at baseline to 394 ms with 5 users and 458 ms with 10 users. This represents an increase of approximately 23.9% from 1 to 5 users and 44.0% from 1 to 10 users. The trend suggests slower average responses in the higher-user run, but the small test sizes do not establish the cause or prove a bottleneck.

4.2 Throughput

Reported throughput was 3.1 requests/second for 1 user, 4.2 requests/second for 5 users and 2.9 requests/second for 10 users. The highest observed throughput was 4.2 requests/second in the 5-user run. Throughput did not increase consistently as the user count rose.

4.3 Error Rate

All three runs reported an error rate of 0.00%. This means no request errors were recorded in the 60 samples reported in the comparison document. It does not guarantee error-free operation during longer tests or sustained concurrent load.

5. Key Findings

- All 60 recorded requests across the three runs reported a 0.00% error rate.
- Average response time increased from 318 ms to 394 ms and then to 458 ms as the tested user count increased.
- The maximum response time was 1,010 ms in the 5-user run and 958 ms in the 10-user run.
- The highest observed throughput was 4.2 requests/second in the 5-user run.
- Standard deviation increased from 181.44 ms to 210.01 ms and 251.00 ms, showing greater variation in response times in the higher-user runs.
- The 5-user average-bytes value was clipped in the source screenshot and has not been estimated.
- The supplied measurements are preliminary and do not establish the website's maximum capacity.

6. Functional Testing Status and Defect Summary

The supplied Toolshop Functional Testing Report contains functional test cases, execution outcomes and Playwright evidence. This performance report keeps those browser-functional results separate from JMeter's HTTP-level measurements. Functional test outcomes should be reported using the exact status counts from the completed test matrix; no overall pass/fail count is inferred here because a verified consolidated count was not supplied as a single summary in the materials reviewed.

For performance testing, the three JMeter runs reported no sample errors. No specific performance defect can be confirmed from these limited results alone. The response-time increase across runs is an observation for further investigation, not a confirmed defect.

7. Limitations

- The sample sizes were small: 5, 25 and 30 requests, respectively.
- The tests were short exploratory comparisons, not a comprehensive load, stress, endurance or capacity test.
- JMeter measured HTTP request/response behaviour for the homepage. It does not render the Angular application or measure browser-side rendering, layout or interaction latency.
- Network variability and server conditions may influence measured response times and throughput.
- Percentile values were visible only for the 10-user run, so percentile comparisons across all three runs are not possible.
- The 5-user average-bytes value was not fully visible and is therefore recorded as “Not fully visible”.
- The public demo website may change; results apply to the conditions and time of the recorded runs.

8. Recommendations

- Repeat the baseline test with a larger sample count to obtain a more stable response-time estimate.
- If further multi-user testing is required, define a controlled ramp-up and duration and capture equivalent metrics for each run.
- Use a locally hosted copy for stress, endurance or volume testing rather than sending heavy traffic to the public demo.
- Capture median, P90, P95 and P99 response times for every run so the tail-latency behaviour can be compared consistently.
- Retain the JMeter .jmx test plan, raw result file, Aggregate Report and screenshots as reproducible evidence.
- If response times remain higher under increased concurrency, repeat the test under comparable network conditions before investigating potential bottlenecks.

9. Test Execution Closure

The supplied evidence supports closure of the initial baseline and comparative HTTP performance checks: 60 requests were recorded across the 1-user, 5-user and 10-user runs, and all reported a 0.00% error rate. Average response time increased with the tested user count, while throughput peaked in the 5-user run. These results provide an initial performance snapshot only. The project should not be described as having passed a full load or stress test because such a test is not demonstrated by the evidence.

10. Conclusion

Apache JMeter was used to compare homepage HTTP performance under three exploratory user configurations. The average response times were 318 ms, 394 ms and 458 ms for 1, 5 and 10 users, respectively. All 60 recorded samples reported no errors, while throughput was highest at 4.2 requests/second in the 5-user run. The response-time trend indicates that further controlled testing would be useful, but the small sample sizes and public-demo environment limit the conclusions that can be drawn. The results are therefore treated as preliminary baseline and comparison measurements rather than proof of scalability or maximum capacity.

Appendix A. Evidence Included

- Figure 1 — HTTP Request sampler screenshot.
- Figure 2 — Thread Group configuration screenshot.
- Figure 3 — Aggregate Report screenshot.
- Figure 4 — HTTP Request Defaults screenshot.
- Figure 5 — Average response time chart from the supplied comparison document.
- Figure 6 — Throughput chart from the supplied comparison document.
- Figure 7 — Error-rate chart from the supplied comparison document.